Grain Definition Document

DailyBasket Data Warehouse - Phase 1
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1. What is Grain?

Grain means what ONE row in a fact table stands for. We decide the grain first. Then we choose the dimensions and measures that match that grain. We never mix two grains in one fact table.

2. Grain Statements

Fact Table	Fact Type	Grain Statement
fact_sales	Transaction	One row represents one product purchased in one order transaction (one order line).
fact_inventory_snapshot	Periodic snapshot	One row represents the end-of-day stock level of one product in one region.
fact_promotion_event	Factless	One row represents one product being part of one promotion on one active day.

3. fact_sales

Grain: One row represents one product purchased in one order transaction (one order line).

What makes a row unique: order_id + order_line_number.

Dimensions at this grain: Date, Customer, Product, Region, Channel, Promotion.

Why this grain: It is the lowest level of detail. We can roll it up to day, month, product, category, region or customer. A higher grain (like one row per order) would lose the product detail.

Example rows:

order_id	line	date_key	customer_key	product_key	region_key	channel_key	promotion_key	qty	net_revenue
O-10045	1	20260315	1001	205	3	2	12	2	2700.00
O-10045	2	20260315	1001	311	3	2	0	1	850.00
O-10046	1	20260315	1042	205	1	1	0	1	1500.00

4. fact_inventory_snapshot

Grain: One row represents the end-of-day stock level of one product in one region.

What makes a row unique: date_key + product_key + region_key.

Dimensions at this grain: Date, Product, Region.

Why this grain: Stock changes all the time. A daily snapshot is enough to see low stock, over-stock and slow products. quantity_on_hand is semi-additive: add it across products or regions, but never across days (use the last day or an average).

Example rows:

date_key	product_key	region_key	quantity_on_hand	quantity_received	quantity_sold	reorder_level	stock_value
20260315	205	3	120	50	18	40	168000.00

date_key	product_key	region_key	quantity_on_hand	quantity_received	quantity_sold	reorder_level	stock_value
20260316	205	3	140	50	30	40	196000.00
20260315	311	3	25	0	6	30	17500.00

5. fact_promotion_event

Grain: One row represents one product being part of one promotion on one active day.

What makes a row unique: date_key + product_key + promotion_key.

Dimensions at this grain: Date, Product, Promotion.

Why this grain: The table records that something happened (a product was on promotion that day). There is no number to measure, so it is a factless fact table. We count rows to answer questions.

Example rows:

date_key	product_key	promotion_key
20260315	205	12
20260316	205	12
20260315	311	12

6. Grain Rules We Follow

- One fact table = one grain.
- Every foreign key in a fact table must match the grain.
- Never store totals or sub-totals as rows in a fact table.
- Measures must make sense at the grain (for example quantity per order line).
- No row can repeat the same combination of its unique key.